

CLEMENT TOH

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EDUCATION

National University of Singapore (NUS)

- Bachelor of Engineering (Mechanical Engineering) with Honours
- Relevant Coursework: Thermodynamics, Circuit Analysis, Additive Manufacturing

Singapore
Expected May 2028
GPA: 4.19 / 5.00

Singapore Polytechnic (SP)

- Diploma in Mechanical Engineering (with Merit)

April 2023
GPA: 3.86 / 4.00

TECHNICAL SKILLS

- CAD & Manufacturing:** SolidWorks, Siemens NX, Autodesk Inventor, Fusion 360, DFM, BOM, Additive Manufacturing (LPBF & FDM)
- Data & Software:** Python (Test Automation, Data Pipelines), C++, InfluxDB, Grafana, MATLAB, PlatformIO, GitLab
- Systems Integration & Embedded:** M5Stack/ IIoT Edge Devices, Arduino, PLCs, EPL (Ethernet POWERLINK) Network Analysis, Electro-Mechanical Troubleshooting, Hardware – Software Integration
- Test, Validation & Instrumentation:** EVT, Root Cause Analysis, NI DAQ (NI-9236), Oscilloscopes, Strain Gauges, Acoustic Emission Sensors, Optical QC, Class 4 Laser Safety & QC

PROFESSIONAL EXPERIENCE

JD Union Pte. Ltd.

Mechatronics Engineering Intern

Singapore
June 2026 – August 2026

- Prevented Optical Failures:** Executed Engineering Validation Testing (EVT) and optical microscopy on Class 4 industrial lasers (LUXINAR CO2), inspecting Quartz Block Head (QBH) fiber optics to prevent catastrophic window failures and eliminate optic downtime.
- Eliminated Test Distortions:** Designed a >250kg 5-tier mobile test workstation in SolidWorks using 4040 aluminium extrusion, dampening mechanical vibrations during sub-micron laser, ensuring DFM compliance for CNC fabrication.
- Optimized Thermal Dissipation:** Commissioned and integrated laser subsystems (IPG fiber laser, chiller, NANIO power supply), routing cable and component layouts to prevent thermal degradation within an enclosed chassis.

Advanced Remanufacturing and Technology Centre (A*STAR)

Systems Integration & Development Engineering Intern

Singapore
September 2022 – April 2023

- Preserved Chamber Integrity:** Engineered custom 13mm feedthrough ports and 3D-printed PCB holders (Siemens NX) to route 12 sensor channels into an EOS M 290 printer, maintaining a sealed inert-gas atmosphere.
- Eliminated Ignition Risks:** Integrated a 200mA fast-acting fuse into the active sensor signal path, eliminating electrical spark risks in a reactive metal powder environment.
- Isolate Micro-Deviations in Real-Time:** Built an end-to-end IIoT telemetry pipeline (12-ch NI DAQ, M5Stack, C++, InfluxDB, Grafana) to aggregate multi-parameter process metrics, successfully isolating powder-bed surface defects.
- Boosted Efficiency by 50%:** Developed Python scripts to tap and decode Ethernet POWERLINK traffic over a 30.6-hour print, implementing data-reduction algorithms cutting 50% overhead, retaining resolution to trace recoater faults.

Singapore Armed Forces (Armour Simulation Wing)

Systems Operator (National Service)

Singapore
May 2023 – May 2025

- Maximized Operational Uptime:** Maintained and debugged networked military simulation hardware alongside defense firms, resolving system faults to achieve operational stability across synchronized training cabins.

TECHNICAL PROJECT

3D Printer Thermal & Hardware Optimization (Ender 3 V3 SE)

June 2024

- Eliminated Part Warpage:** Redesigned print-head shroud in Fusion 360 and integrated high-performance dual blowers, optimizing airflow to resolve thermal inconsistencies and eliminate geometric distortion on steep overhangs.

LEADERSHIP & COMMUNITY INVOLVEMENT

Boon Lay Youth Network

Grassroots Leader

Singapore
January 2022 – December 2023

- Scaled Tutoring Operations:** Directed online academic support programs for underprivileged youth, managing 20 volunteer educators across 20 concurrent breakout sessions to maintain seamless tutoring delivery.